

Use case CUD Product

Use Case: Create Product

1. Use case code

UC004-Create

2. Brief Description:

This use case describes the interaction between the Product Manager and the AIMS system when the actor wants to add a new product to the system. The Product Manager can add an unlimited number of products per day , but only one product at a time. The system validates the input data and ensures that all business rules are satisfied before completing the operation.

3. Actors:

Product Manager

4. Preconditions:

- The Product Manager needs to log in to access the features corresponding to their roles.
- The Product Manager is currently on the product management screen.

5. Basic Flow of Events:

1. The Product Manager accesses the product management screen.
2. The Product Manager chooses to add a new product.
3. The system displays a product creation form.
4. The Product Manager enters the essential core information and physical attributes of the product.
5. The Product Manager selects the product category (books, newspapers, CDs, or DVDs) and provides further details depending on the type of product.
6. The Product Manager submits the request.
7. The system validates the input data to ensure it complies with the rules.
8. The system stores the new product information in the database.
9. The software stores a history of products addition.
10. The system displays a success notification.
11. The use case ends.

6. Alternative flows:

- **Invalid Input:** The software will notify the product manager if any operation is invalid. For example, when the input does not comply with the rules or the date format is incorrect. Resume at Step 4 or 5.
- **Price Constraint Violation:** The price of the product must always be between 30% and 150% of the product's original value to avoid price inflation or undercutting. If violated, the system displays an error message. Resume at Step 4.

7. Input data:

No	Data fields	Description	Mandatory	Valid condition	Example
1	Barcode	Core detail: Unique identifier	Yes	Must not duplicate existing barcode	893523451234
2	Title	Core detail: Product title	Yes	Non-empty text	One Piece Vol.1
3	Category	Core detail: books, newspapers, CDs, DVDs	Yes	Must belong to supported types	Book
4	Original value	Core detail: Original value without a 10% value-added tax (VAT)	Yes	Must be positive number	100000
5	Current price	Core detail: Current price without a 10% value-added tax (VAT)	Yes	Between 30% and 150% of original value	120000
6	Description	General description (e.g., new or used condition, primary color, or return condition)	Yes	Text	Brand new, sealed
7	Dimensions	Physical attributes: Height, width, length	Yes	Valid measurement	20x15x5 cm
8	Weight	Physical attribute	Yes	Positive number	0.5 kg
9	Book details	Author(s), cover type (paperback or hardcover), publisher, publication date. Additional: number	Yes (if Book)	Follow product rules	Author: Eiichiro Oda, Cover: Paperback, Publisher: Shueisha

		of pages, language, genre.			
10	Newspaper details	Editor-in-chief, publisher, publication date. Additional: issue number, publication frequency, ISSN, language, sections.	Yes (if Newspaper)	Follow product rules	Editor: John Doe, Frequency: Daily, Section: Sports
11	CD details	Artists, record labels, tracks list (title and length), genre. Additional: release date.	Yes (if CD)	Follow product rules	Artist: The Beatles, Genre: Rock
12	DVD details	Disc type (Blu-ray, HD-DVD), director, runtime, studio, language, subtitles. Additional: release date, genre.	Yes (if DVD)	Follow product rules	Type: Blu-ray, Director: James Cameron, Runtime: 192 mins

8. Output data:

No	Data fields	Description	Display format	Example
1	Operation result	Notification of successful product addition.	Text message	Product added successfully
2	Error / Validation message	Notification if the input does not comply with the rules or date format is incorrect.	Text message	Invalid date format / Current price must be at least 30% of original value

9. Postconditions:

- The product information is successfully created in the system.
- The software stores a history of products addition.
- The Product Manager can continue adding other products (unlimited per day).

Use Case: Update Product

1. Use case code

UC004-Update

2. Brief Description:

This use case describes the interaction between the Product Manager and the AIMS system when the actor wants to edit the information of an existing product. The system validates the input data and ensures that all business rules are satisfied before completing the operation.

3. Actors:

Product Manager

4. Preconditions

- The Product Manager has logged into the system to access the features corresponding to their role.
- The selected product must already exist in the system.

5. Basic Flow of Events

1. The Product Manager accesses the product management screen.
2. The system displays the list of products.
3. The Product Manager chooses to edit the information of an existing product.
4. The Product Manager selects only one product at a time to edit.
5. The system retrieves and displays the current product information.
6. The Product Manager modifies the required information (including stock adjustment and explicit reason if any) and submits the changes.
7. The system validates the input data to ensure it complies with the rules.
8. The system saves the updated product information in the database.
9. The software stores a history of product editing.
10. The system displays a success notification.
11. The use case ends.

6. Alternative flows

- **Invalid Input:** The software will notify the product manager if any operation is invalid. For example, when the input does not comply with the rules or the date format is incorrect. Resume at Step 6.
- **Price Constraint Violation:** The price of the product must always be between 30% and 150% of the product's original value to avoid price inflation or undercutting. If violated, the system displays an error message. Resume at Step 6.

7. Input data

No	Data fields	Description	Mandatory	Valid condition	Example
1	Title	Product title	Yes	Non-empty text	Dragon Ball
2	Category	Product category (books, newspapers, CDs, DVDs)	Yes	Must belong to supported types	newspaper
3	Original value	Original value without a 10% value-added tax (VAT)	Yes	Must be positive number	200000
4	Current price	Current selling price without a 10% value-added tax (VAT)	Yes	Between 30% and 150% of original value	210000
5	Description	General description (e.g., new or used condition, primary color, or return condition)	Yes	Text	slight wear
6	Dimensions	Height, width, length	Yes	Valid measurement	20x15x15 cm
7	Weight	Physical attribute	Yes	Positive number	0.6 kg
8	Quantity in stock	Number of items in stock	Yes	Integer ≥ 0	120
9	Reason for stock change	Explicitly recorded reason when stock needs to be adjusted manually	Yes (if stock is changed manually)	Non-empty text	Removed 2 damaged items
10	Product type specific info	Details depending on the type of product	Yes	Must follow product type rules	Artist: The Beatles, Genre: Rock

8. Output data:

No	Data fields	Description	Display format	Example
1	Operation result	Notification of successful product update.	Text message	Product updated successfully
2	Error / Validation message	Notification explaining why an operation is invalid.	Text message	The price must be between 30% and 150% of the original value

3	Updated product information	Updated product data displayed after operation.	Structured text	Quantity in stock: 150
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9. Postconditions:

- The product information is successfully updated in the system.
- The software stores a history of product editing.
- The product list is updated to reflect the latest changes.

Use Case: Delete Product

1. Use case code:

UC004-Delete

2. Brief Description:

This use case describes the interaction between the Product Manager and the AIMS system when the actor wants to delete products. The system checks simultaneous deletion limits, daily deletion limits, and stock quantities before performing the deletion or changing the product status to deactivated.

3. Actors:

Product Manager

4. Preconditions:

- The Product Manager needs to log in to access the features corresponding to their roles.
- The Product Manager is currently on the product management screen.
- The selected product(s) must already exist in the system.

5. Basic Flow of Events:

1. The Product Manager accesses the product management screen.
2. The Product Manager selects up to ten products to delete simultaneously.
3. The Product Manager submits the request to delete the selected products.
4. The system validates the daily deletion limit to ensure the manager does not delete more than twenty products per day.
5. For each selected product, the system checks if its stock is equal to zero.

6. Confirming the stock is zero, the system deletes the product(s).
7. The software stores a history of products deletion.
8. The system displays a success notification.
9. The use case ends.

6. Alternative flows:

- **Exceed Simultaneous Limit:** If the manager selects more than ten products at a time, the software will notify the product manager that the operation is invalid. Resume at Step 2.
- **Exceed Daily Limit:** If the deletion request causes the manager to exceed deleting twenty products per day, the system rejects the operation for security reasons and notifies the manager. Use case ends.
- **Stock Greater Than Zero:** If a product's stock is not equal to zero, it cannot be deleted. Instead, its status will be changed to "deactivated," meaning it will no longer be available for sale. Resume at Step 7.

7. Input data

No	Data fields	Description	Mandatory	Valid condition	Example
1	Product ID (Barcode)	Unique identifier of the selected product(s)	Yes	Must exist in the system. Up to ten products may be deleted simultaneously.	893523451234, 893523451235

8. Output data

No	Data fields	Description	Display format	Example
1	Operation result	Notification of successful product deletion or status change.	Text message	Products deleted successfully
2	Error / Validation message	Notification if the operation is invalid (e.g., limit exceeded).	Text message	Cannot delete more than twenty products per day

3	Product status	Status changed to "deactivated" if stock > 0.	Text	Deactivated
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9. Postconditions

- The selected products are deleted from the system, or their status is changed to "deactivated" if their stock was greater than zero.
- The software stores a history of products deletion.